

Assignment 5

Jas Prajapati-110147267

Neel Shah-110157144

Urvesh Shah -110151377

Tejas Soni- 110139054

- **Bidirectional DC-DC Converters**

DC-DC converters are essential components in electric vehicle charging systems (EVCS). These converters can be either unidirectional—such as those used in photovoltaic (PV) systems—or bidirectional, enabling power flow in both directions. A significant application of bidirectional DC-DC converters is vehicle-to-grid (V2G) technology, also referred to as vehicle-grid integration. This approach allows plug-in electric vehicles (PEVs) to supply power back to the grid during peak demand periods, effectively turning them into energy storage units.

By implementing V2G technology, the integration of renewable energy sources into the grid becomes more viable and efficient. Similar to other energy storage systems (ESS), such as batteries and supercapacitors, V2G can help stabilize power fluctuations from renewable sources like PV systems, which may experience sudden output variations due to weather changes.

One real-world demonstration of V2G's potential occurred during the *Zero Emission Mobility to All* (Zem2All) project in Spain, where six bidirectional DC chargers were deployed to evaluate the feasibility and impact of this technology. The collected data highlighted the role of electric mobility in reducing CO₂ emissions. However, a notable challenge of V2G adoption is its effect on battery lifespan, as frequent charge-discharge cycles can accelerate battery degradation.

- **Circuit Components and Topology**

A typical **DAB converter** consists of the following key components:

- **Two Active Bridges (H-Bridge Inverters):**

- **Primary-side H-Bridge:** Connected to the input DC source (e.g., battery or DC bus).
- **Secondary-side H-Bridge:** Connected to the output DC source.
- Both bridges are made of four semiconductor switches (MOSFETs or IGBTs) arranged in an **H-bridge** configuration.

- **High-Frequency Transformer:**

- Provides galvanic isolation.
- Steps up or steps down the voltage depending on the turns ratio.
- Transfers power using a high-frequency AC signal.

- **AC Link and Resonant Inductor (Leakage Inductance):**

- An inductor (either external or the transformer's leakage inductance) facilitates power transfer between bridges by controlling the phase shift of switching signals.

- **Control Circuitry:**

- Implements **phase shift modulation** to regulate power flow.
- Adjusts phase difference between the primary and secondary bridge voltages.

- **Working Principle of DAB Converter**

The power transfer mechanism is based on **phase shift control** of the AC voltage generated by the bridges:

Step 1: DC to High-Frequency AC Conversion:

- The primary H-bridge converts the input DC voltage into a high-frequency square wave.
- The switching signals are controlled to generate an alternating waveform.

Step 2: Power Transfer via Transformer:

- The high-frequency AC passes through the transformer and inductance.
- The magnitude and direction of power flow depend on the **phase shift (ϕ)** between the primary and secondary bridges.

Step 3: AC to DC Conversion at the Output:

- The secondary H-bridge converts the AC waveform back into DC.

- The resulting output voltage depends on the **turns ratio** and the phase shift.

Step 4: Bi-Directional Operation:

- When the phase shift (ϕ) is **positive**, power flows from the primary side (input) to the secondary side (output).
- When the phase shift (ϕ) is **negative**, power flows in the opposite direction, making the converter **bi-directional**.

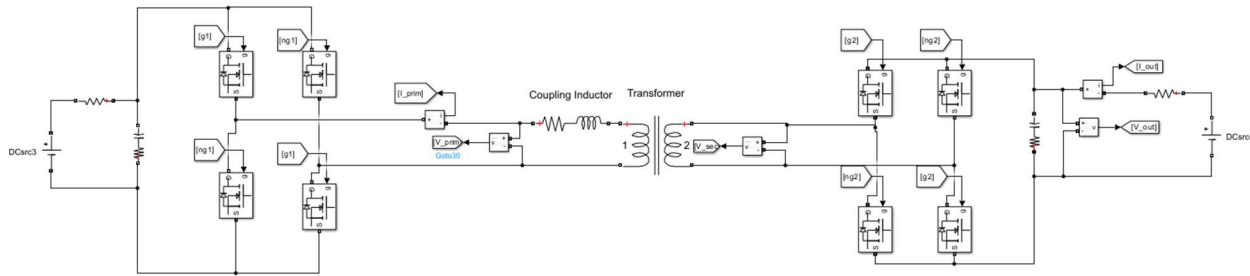


Figure 1 Circuit

• Circuit Diagram Representation

A simplified circuit diagram of a **Bi-Directional DAB Converter** consists of:

- **Primary H-Bridge** (Switches: S1, S2, S3, S4)
- **High-Frequency Transformer** (with leakage inductance L)
- **Secondary H-Bridge** (Switches: S5, S6, S7, S8)
- **DC Sources** (V1 and V2)
- **Control Unit** (for phase shift modulation)
- **Mathematical Model**

$$P = \frac{V_{HV} N V_{LV} \phi (\pi - |\phi|)}{2 \pi^2 F_{SW} L}$$

- V_{HV} is the DC link voltage of the high voltage bridge, in Volts
- N is the primary to secondary transformer voltage ratio
- V_{LV} is the DC link voltage of the low voltage bridge, in Volts
- ϕ is the phase shift between the two square waves of the bridges, in Radians
- F_{SW} is the PWM switching frequency, in Hertz
- L is the total inductance (coupling inductor + total transformer leakage reactance seen from the primary) between the two bridges, in Henry.

• Applications in the Automotive Context

The DAB converter is commonly used in electric vehicles (EVs) and hybrid electric vehicles (HEVs) because it can transfer power efficiently in both directions and provides electrical isolation through the transformer.

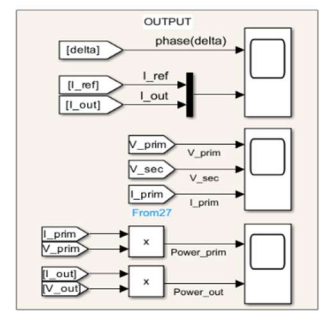
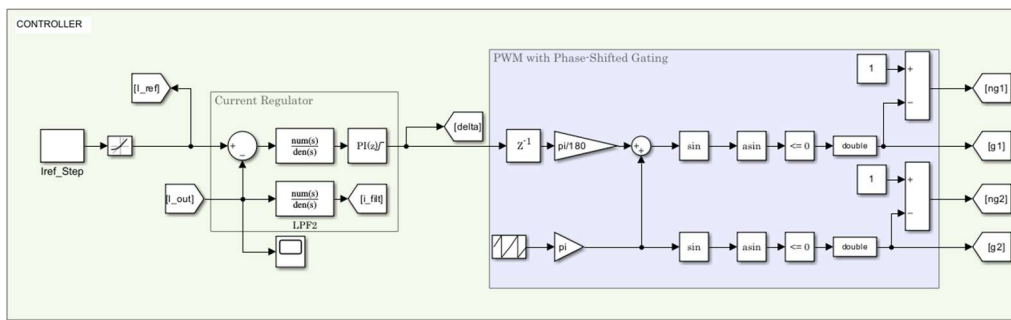
Some important applications in the automotive field include:

Onboard Chargers: Allowing power to flow in both directions for charging and regenerative braking.

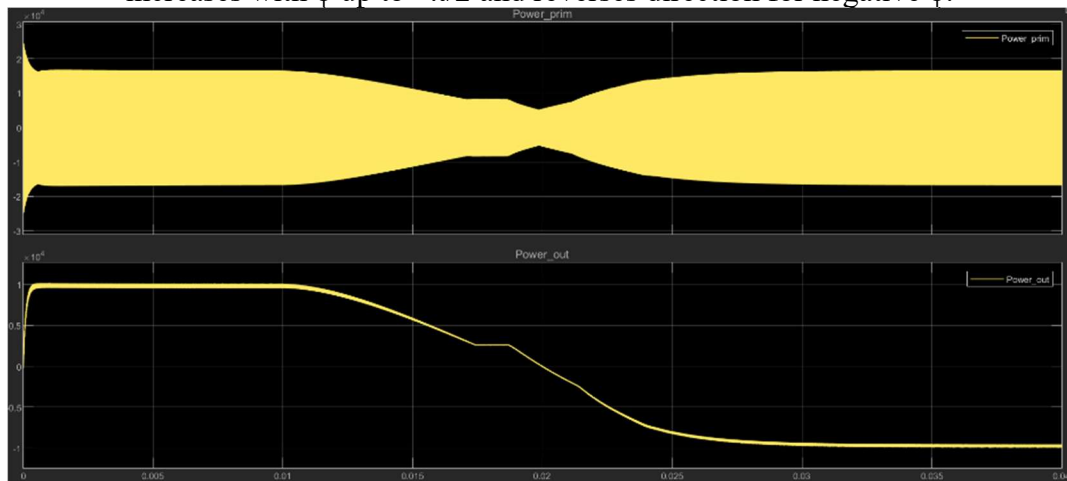
Battery Management Systems: Managing power between batteries and other vehicle systems.

DC Fast Charging Stations: Converting power between DC grids and EV batteries efficiently.

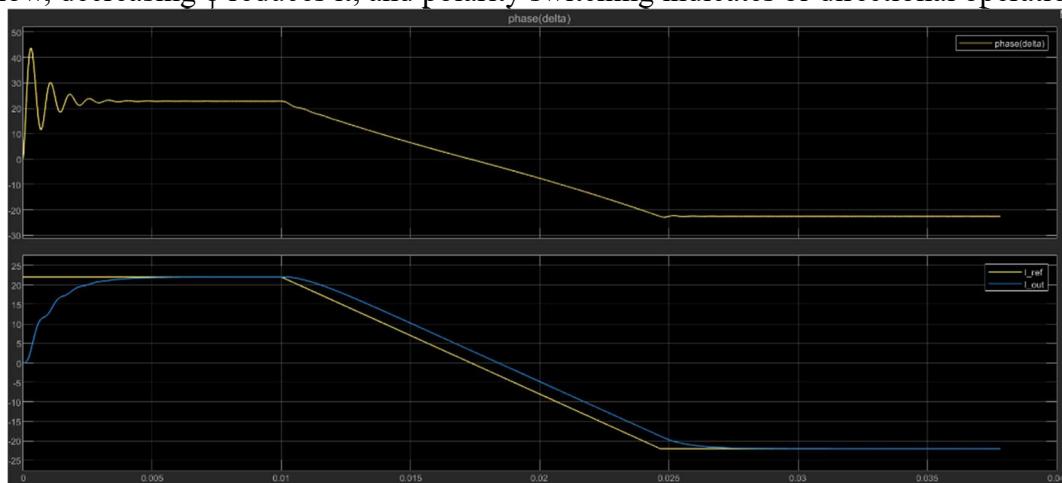
The DAB converter's compact design, efficiency, and ability to handle bidirectional power flow make it ideal for modern electric vehicle systems.



- Power Transfer Graph:** Shows a parabolic relationship between power and phase shift (ϕ), where power increases with ϕ up to $\pm\pi/2$ and reverses direction for negative ϕ .



- Phase Shift Graph:** Displays the variation of phase shift over time, where increasing ϕ boosts power flow, decreasing ϕ reduces it, and polarity switching indicates bi-directional operation.



- Current Reference & Output Current Graph:** Illustrates how the actual output current (I_{out}) follows the reference current (I_{ref}), with possible minor deviations due to control dynamics and transient oscillations.

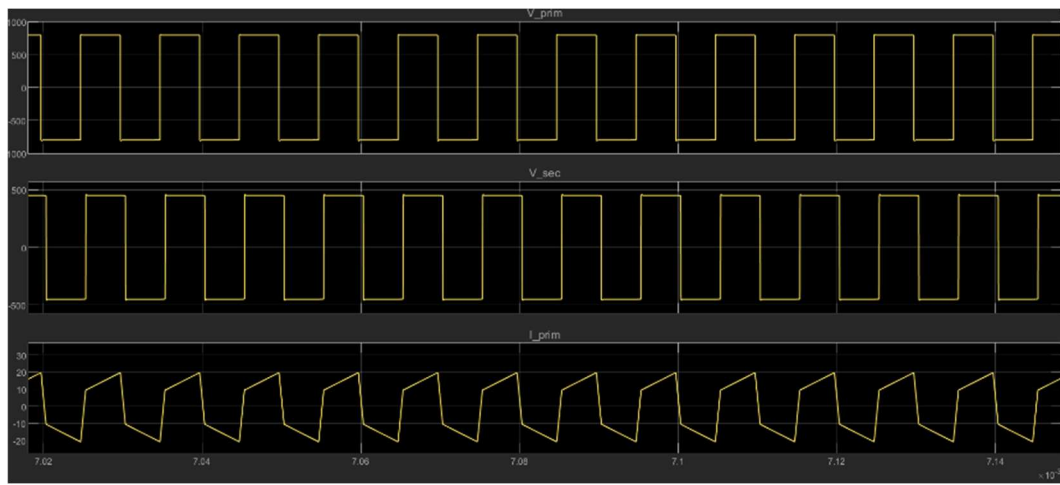


Figure 6 Input Voltage, Current and Output Voltage

• References

- [1] Dual active bridge converter DC/DC converter, <https://www.mathworks.com/help/sps/ug/power-converters-dab-1ph.html> (accessed Mar. 31, 2025).
- [2] (PDF) design and control of Bidirectional Dual Active Bridge DC-DC converter for Dynamic Electric Vehicle Charging and vehicle-to-grid operations, https://www.researchgate.net/publication/361681772_Design_and_Control_of_Bidirectional_Dual_Active_Bridge_DC-DC_Converter_for_Dynamic_Electric_Vehicle_Charging_and_Vehicle-to-Grid_Operations (accessed Apr. 1, 2025).
- [3] Explicit discrete modelling of bidirectional dual active bridge ..., <https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/iet-pel.2020.0293> (accessed Apr. 1, 2025).